CONSTRUCTING R-INVARIANT OPENS IN A GROUPOID SCHEME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-pre-equivalence-equivalence-relation-points (canonical tag 022Q). cited at proof lines 3826-3828. Describes the equivalence relation on points of U.
- Lemma lemma-groupoid-pre-equivalence (canonical tag 0232). cited at proof lines 3826-3828. The groupoid scheme gives a pre-equivalence relation.

Source reference. Original lemma 022Q: sources/stacks-chow/tags.txt, source lines 2715–2715. Cited at proof lines 3826-3828 for part (1): identifies t(s^-1(W)) with the points equivalent to W; load-bearing.

Source reference. Original lemma 0232: sources/stacks-chow/tags.txt, source lines 2727–2727. Cited at proof lines 3826-3828 together with the previous lemma for part (1); load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S.*

(1) *For any subset $W \subset U$ the subset $t(s^{-1}(W))$ is set-theoretically R-invariant.*
(2) *If s and t are open, then for every open $W \subset U$ the open $t(s^{-1}(W))$ is an R-invariant open subscheme.*
(3) *If s and t are open and quasi-compact, then U has an open covering consisting of R-invariant quasi-compact open subschemes.*

Proof. Part (1) follows from Lemmas 022Q and 0232, namely, $t(s^{-1}(W))$ is the set of points of U equivalent to a point of W. Next, assume s and t open and $W \subset U$ open. Since s is open the set $W' = t(s^{-1}(W))$ is an open subset of U. Finally, assume that s, t are both open and quasi-compact. Then, if $W \subset U$ is a quasi-compact open, then also $W' = t(s^{-1}(W))$ is a quasi-compact open, and invariant by the discussion above. Letting W range over all affine opens of U we see (3). □

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